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## Push

(floor-plane for flat objects)



### Problem

Pick-and-place fails on thin/flat objects (<2 cm tall) — the gripper can't get under them, or the grasp slips on every attempt.



### When to Apply

- Object height < 2 cm
- Pick-and-place already failed  $\geq 1$  time
- Target pose is reachable by a horizontal push



### Strategy

Push along the floor plane instead of grasping. The simulator's contact model handles Z-penetration; the action reduces to a 2D translation on the support surface.

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## Knob Turning

(yaw-controlled rotation)

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## Drawer Opening

(validated top-drawer)

...



## Push

```
def push_object(obj_pos, target_xy):
    direction = (target_xy - obj_pos[:2]) /
np.linalg.norm(target_xy - obj_pos[:2])
    pre_push = obj_pos[:2] - direction * 0.10
    post_push = target_xy

    move_to([pre_push[0], pre_push[1], obj_pos[2]])
    move_to([post_push[0], post_push[1], obj_pos[2]])
    # linear push
```



## Motion Primitive